EXPERIMENT NO.8

Name: Vaishnal Mali
Class:D20A
Roll No:32
Batch:B

Aim: To implement the Blockchain platform Ganache and deploy a simple Solidity smart contract using Metamask and Remix IDE.

Theory:

1. Introduction to Blockchain

Blockchain is a **distributed, decentralized digital ledger** that records transactions across multiple nodes in a network. Each transaction is stored in a block, and blocks are linked together using cryptographic hashes, forming a secure and immutable chain.

Key features:

- **Decentralization** – No central authority controls the system
- **Immutability** – Data cannot be altered once recorded
- **Transparency** – Transactions are visible to all participants
- **Security** – Uses cryptographic techniques

2. What is Ganache?

Ganache is a **personal blockchain simulator** used for developing, testing, and debugging Ethereum-based decentralized applications (DApps).

It is part of the **Truffle Suite**, which provides tools for smart contract development.

Ganache creates a **local blockchain environment** that mimics the behavior of the real Ethereum network.

3. Features of Ganache

a. Local Blockchain

Ganache runs a blockchain locally on your system, allowing developers to test without using real cryptocurrency.

b. Pre-funded Accounts

- Provides multiple accounts with preloaded Ether
- Useful for testing transactions without cost

c. Instant Mining

- Transactions are mined instantly
- No waiting time like real networks

d. Detailed Transaction Logs

- Displays:
    - Transaction hash
    - Gas usage
    - Block number
    - Events

e. Smart Contract Testing

Supports deployment and testing of smart contracts written in Solidity.

f. Customizable Settings

- Gas limit
- Gas price
- Number of accounts
- Network ID

4. Architecture of Ganache

Ganache simulates the Ethereum architecture:

a. Accounts Layer

- Contains multiple Ethereum accounts
- Each account has:
  - Address
  - Private key
  - Balance

b. Blockchain Layer

- Stores blocks and transactions
- Each block contains:
  - Block number
  - Timestamp
  - Transactions

c. Virtual Machine

Ganache uses the **Ethereum Virtual Machine** to execute smart contracts.

d. RPC Server

- Provides JSON-RPC API
- Allows interaction using tools like Web3.js

5. Working of Ganache

1. Ganache starts a local blockchain network
2. It generates multiple accounts with Ether
3. Developers deploy smart contracts
4. Transactions are executed
5. Ganache mines blocks instantly
6. Results are displayed in logs

6. Advantages of Using Ganache

- No real Ether required
- Faster testing (instant mining)
- Easy debugging with detailed logs
- Safe environment (no risk of real loss)
- Ideal for beginners and developers

7. Limitations of Ganache

- Not a real decentralized network
- Does not simulate real-world latency
- Limited scalability testing
- Not suitable for production deployment

8. Applications of Ganache

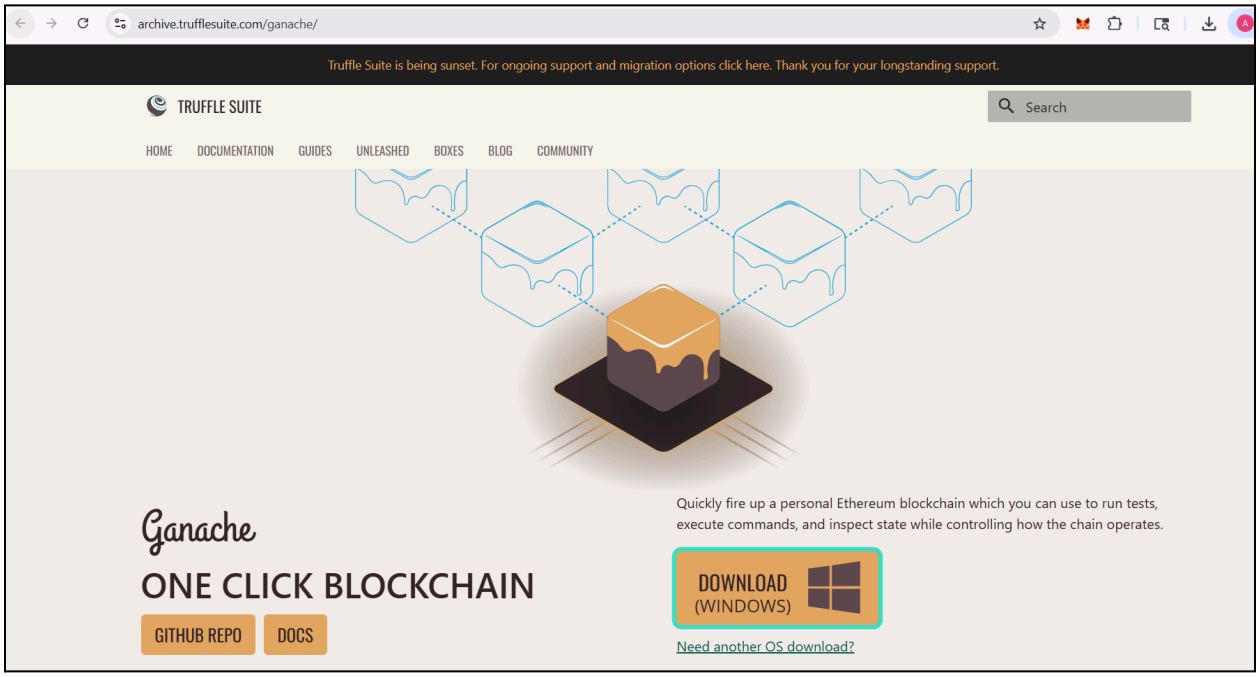
- Smart contract development
- Testing decentralized applications (DApps)
- Learning Ethereum blockchain concepts

- Debugging transactions

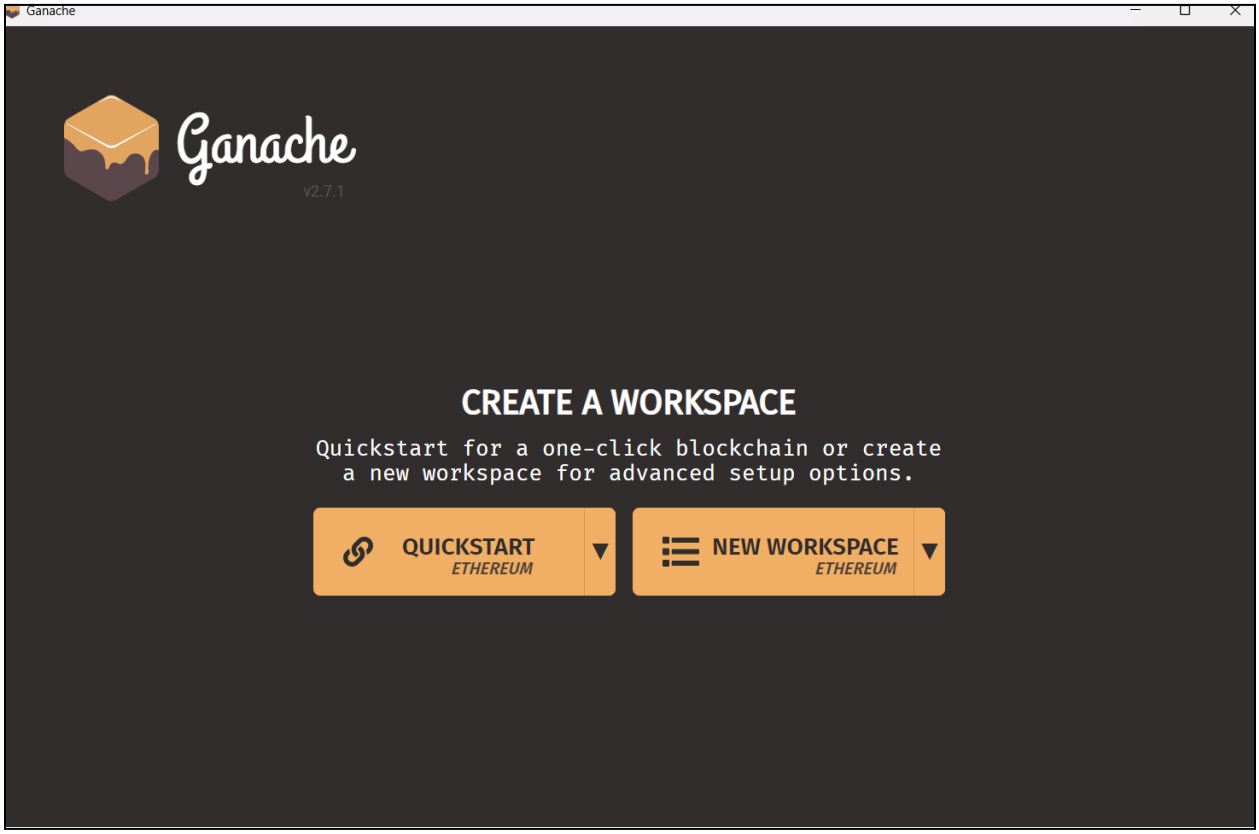
Implementation:-

1. Set Up Ganache (The Local Blockchain)

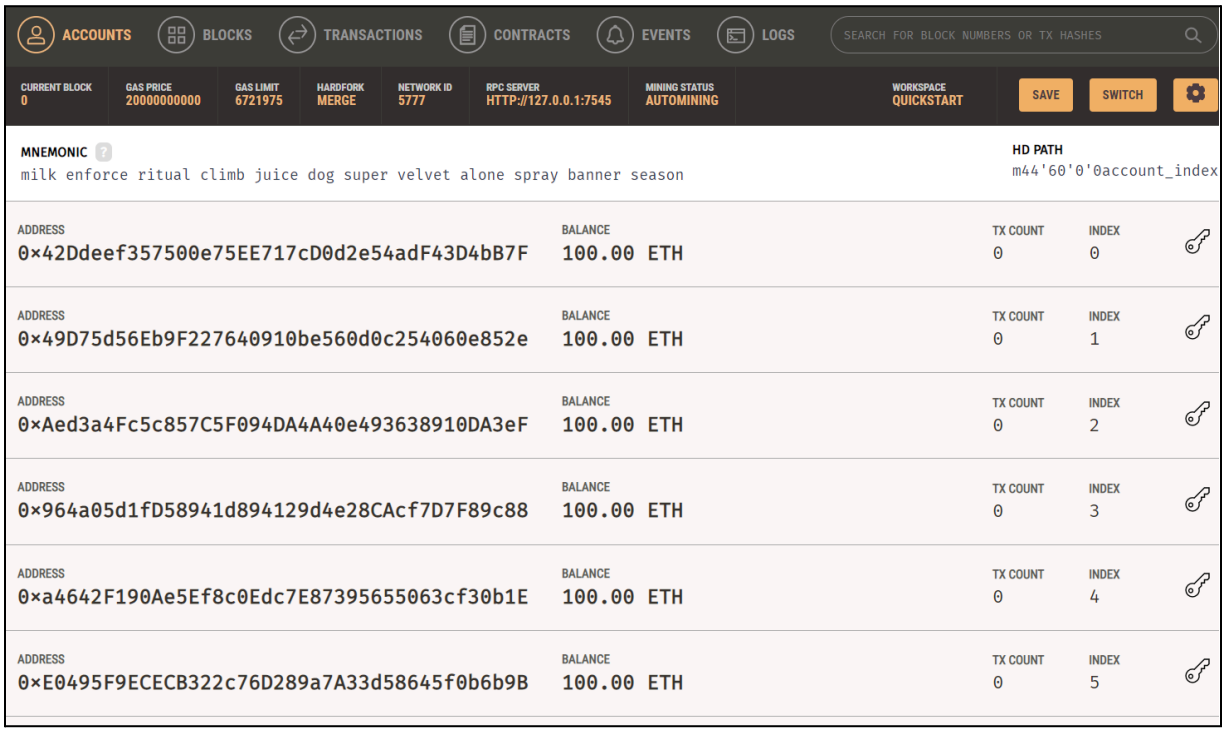
1. **Download & Install**: Go to the [Truffle Suite website]() and install Ganache.



2. **Launch**: Open Ganache and select **"Quickstart"**.



3. **Identify Details**: You will see a list of 10 addresses, each with 100 ETH.
    - Look at the top bar for **RPC Server**. It usually says `HTTP://127.0.0.1:7545`.
    - Look at the **Network ID**. It is usually `5777`.

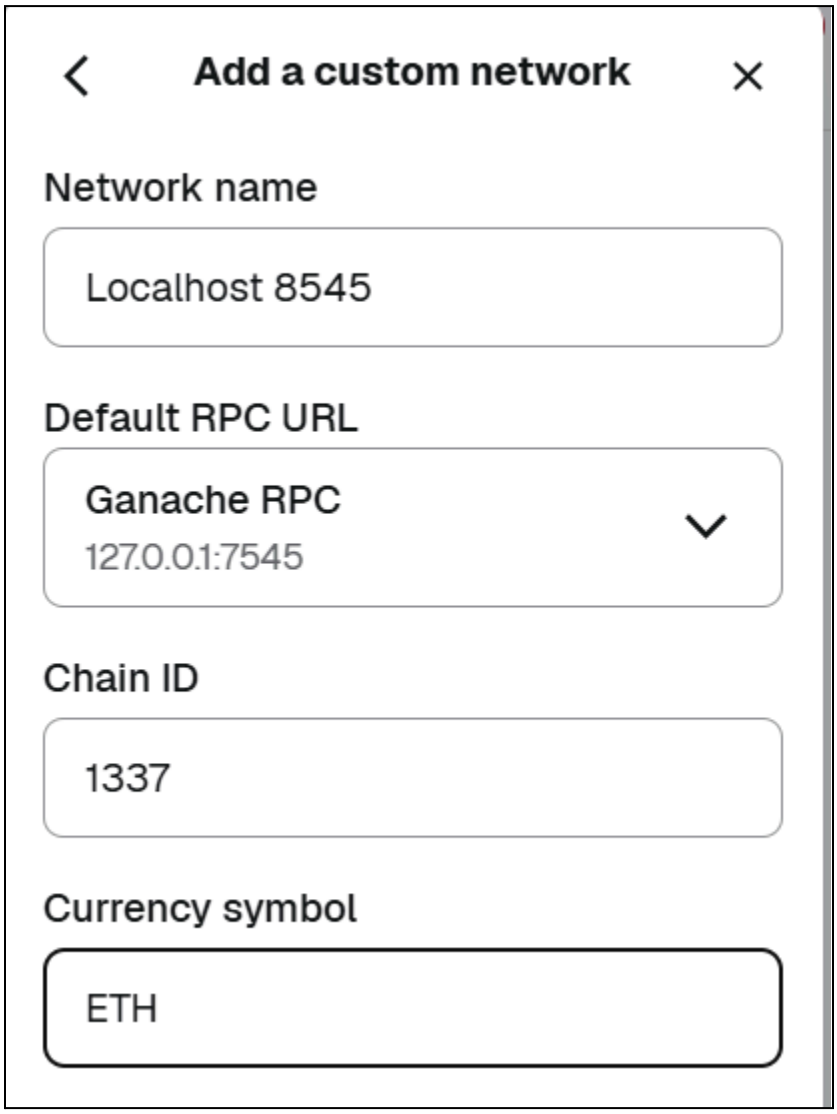


2. Connect Ganache to MetaMask

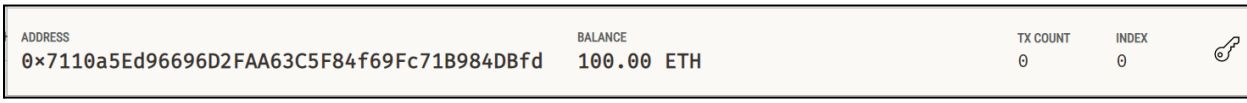
MetaMask doesn't know about your local Ganache "server" yet, so we have to add it manually.

1. Open **MetaMask** -> Click the **Network Dropdown** (top left) -> **Add Network**.
2. Select **"Add a network manually"** at the bottom.
3. Enter these details:
    - **Network Name**: Localhost 8545
    - **New RPC URL**: `http://127.0.0.1:7545`
    - **Chain ID**: `1377`

- **Currency Symbol**: ETH



4. **Import Account**:
   - In Ganache, click the **Key Icon** on the far right of the first address.



   - Copy the **Private Key**.

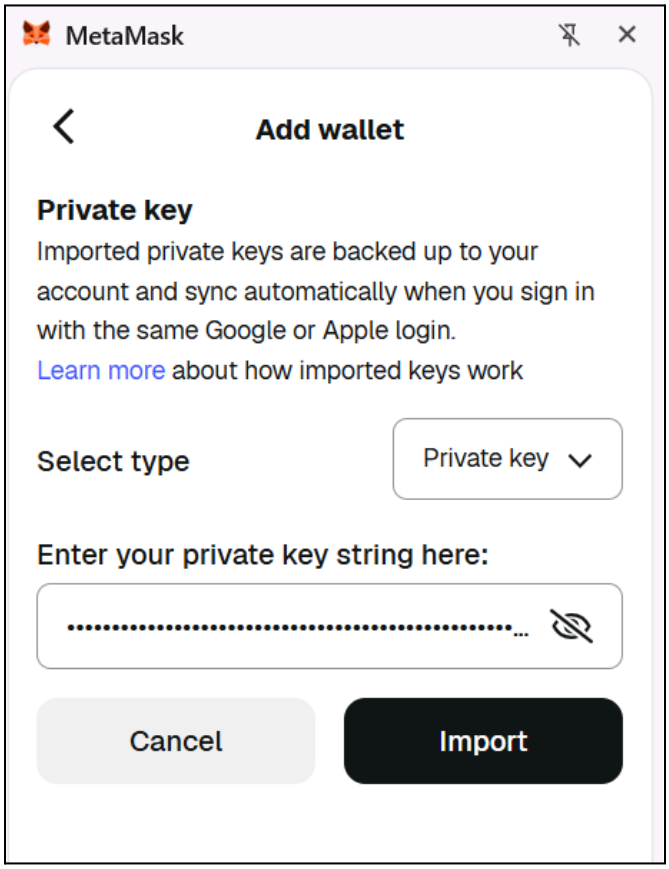
ACCOUNT ADDRESS

0×7110a5Ed96696D2FAA63C5F84f69Fc71B984DBfd

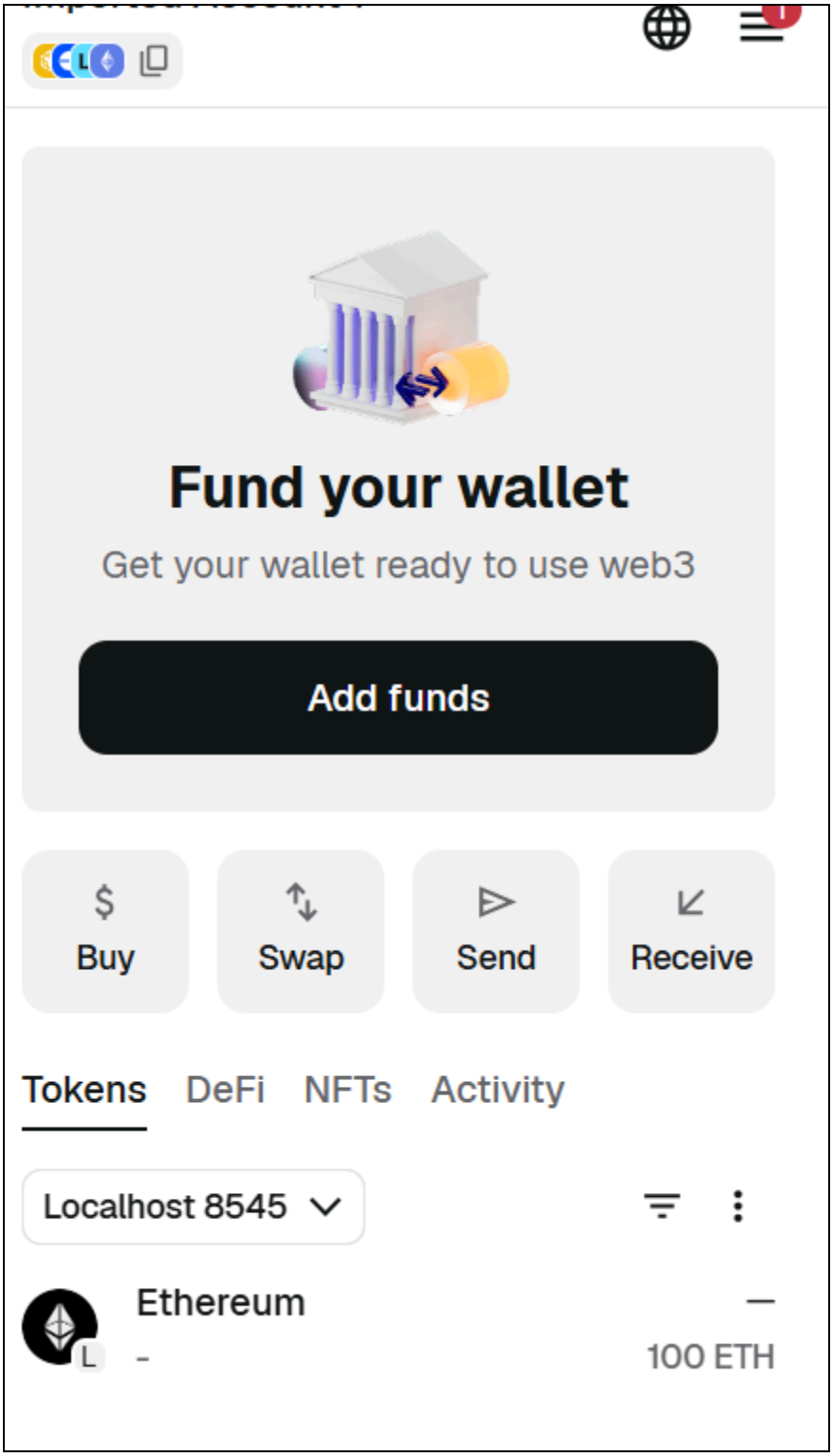
PRIVATE KEY

0×f4a7a11cb7e2aa839bae7d439672da8684ee9d061f2b809eb0c0da4700e845

- In MetaMask, click the Account Circle -> **Import Account** -> Paste the Private Key.

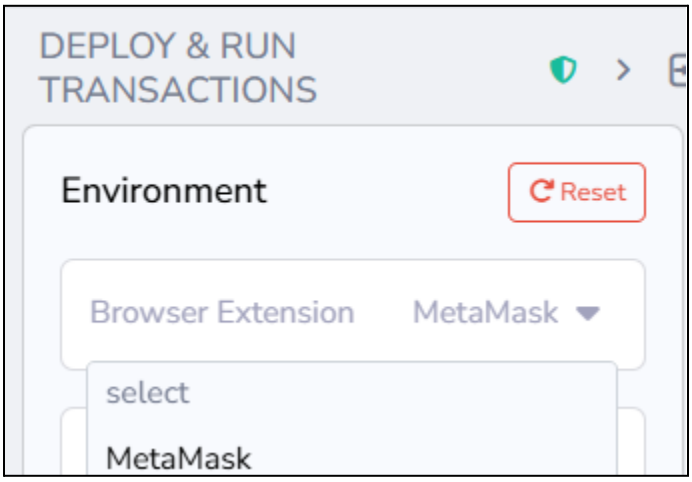


- You should now see **100 ETH** in MetaMask!

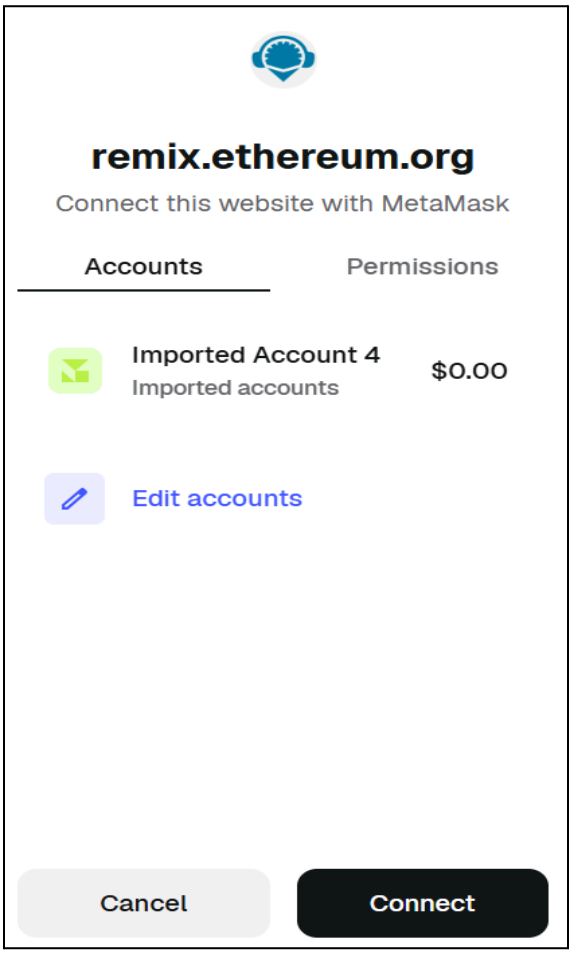


3. Connect Remix to MetaMask (Injected Provider)

1. In Remix, go to the **Deploy & Run Transactions** tab.
2. Set **Environment** to **Injected Provider - MetaMask**.

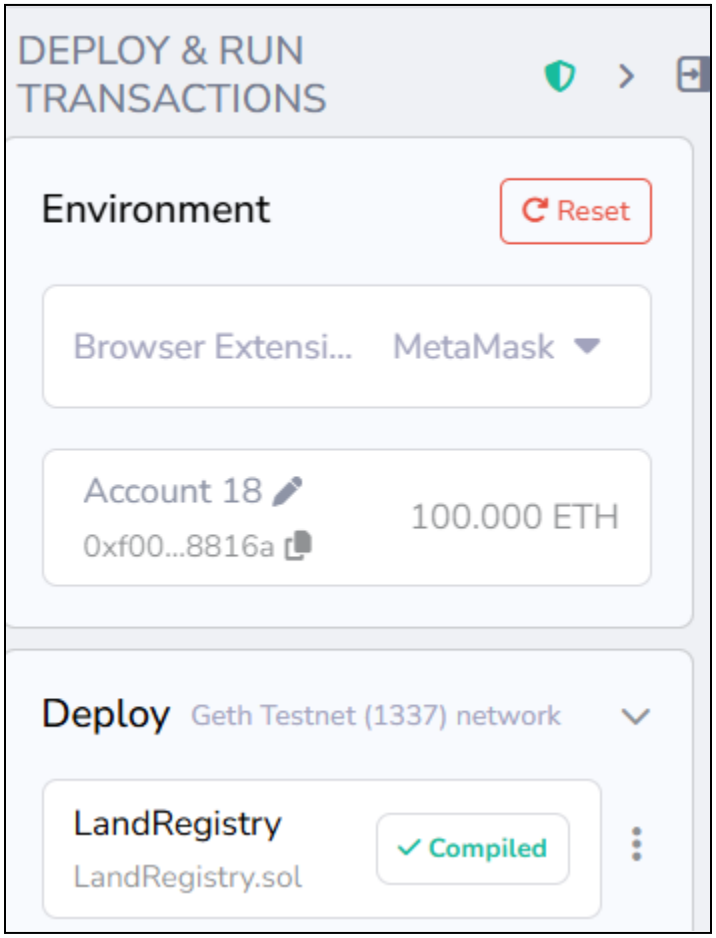


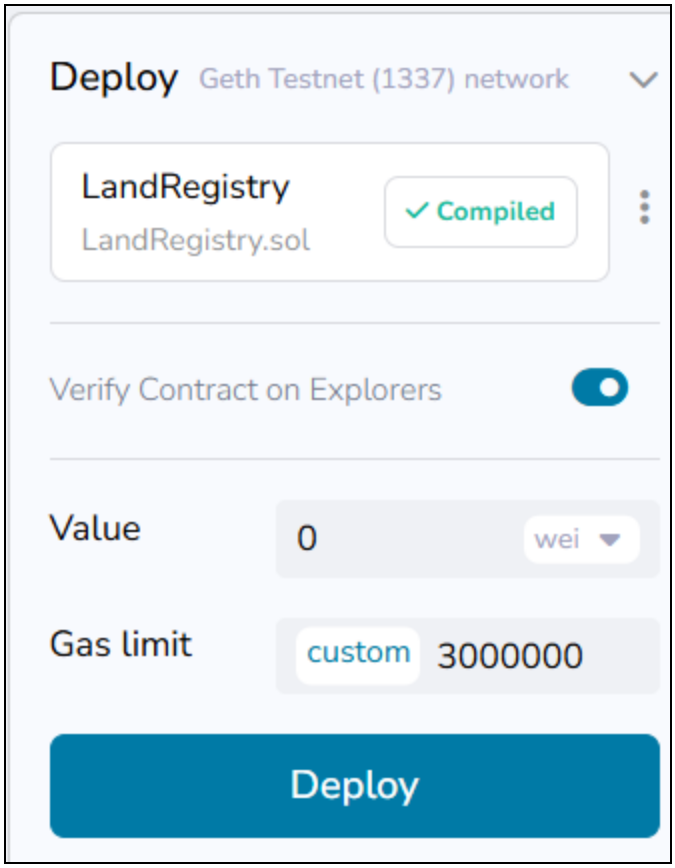
3. MetaMask will ask to connect to "Ganache Local." Click **Confirm**.



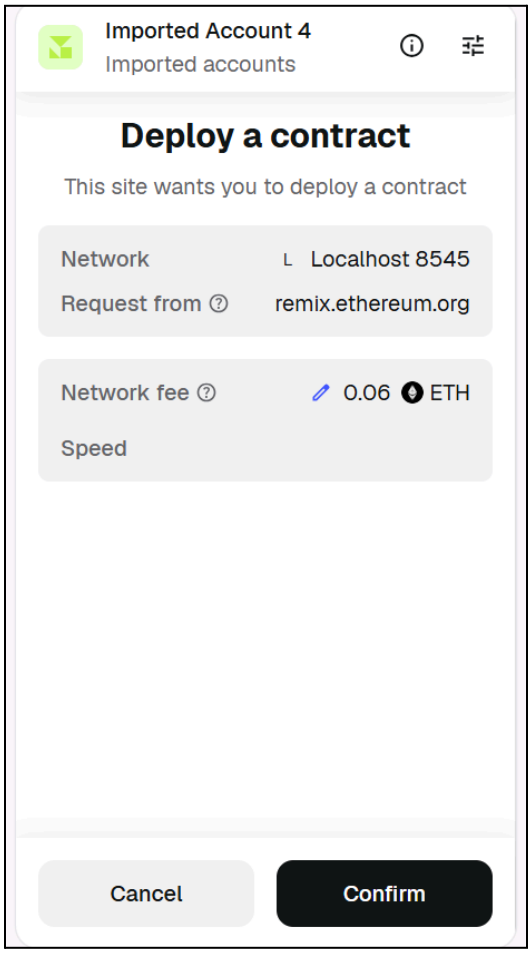
4. Compile and Deploy

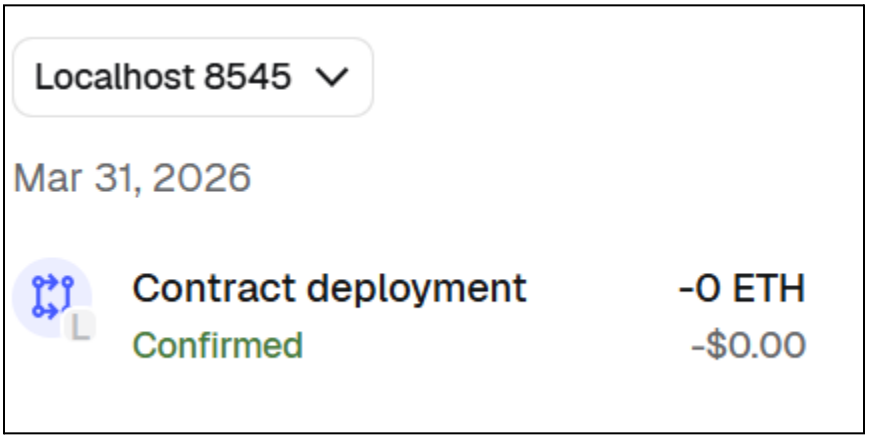
1. Compile your `LandRegistry.sol`.
2. In the Deploy tab, click **Deploy**.




Deploy
Geth Testnet (1337) network
LandRegistry
LandRegistry.sol
Compiled
Verify Contract on Explorers
Value
0
wei
Gas limit
custom
3000000
Deploy

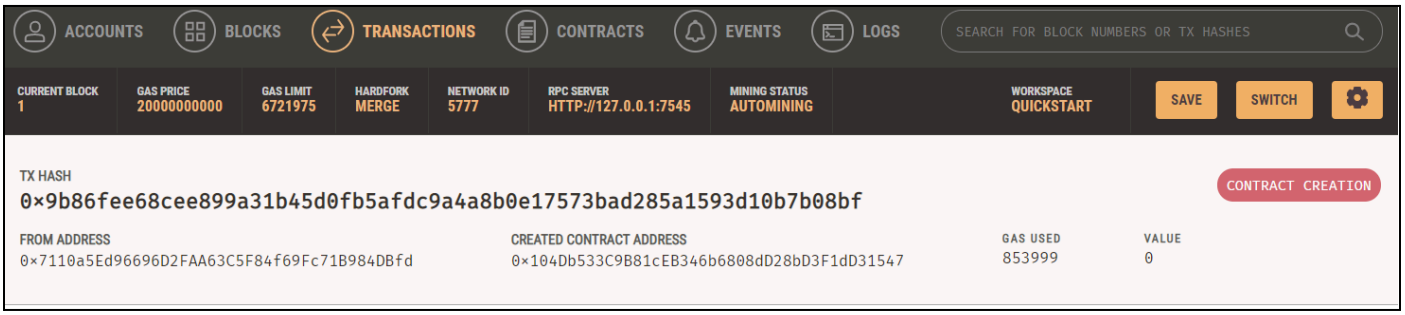
3. MetaMask will pop up. Since this is your local Ganache, the transaction will be **instant** and the gas cost will be deducted from your 100 ETH.

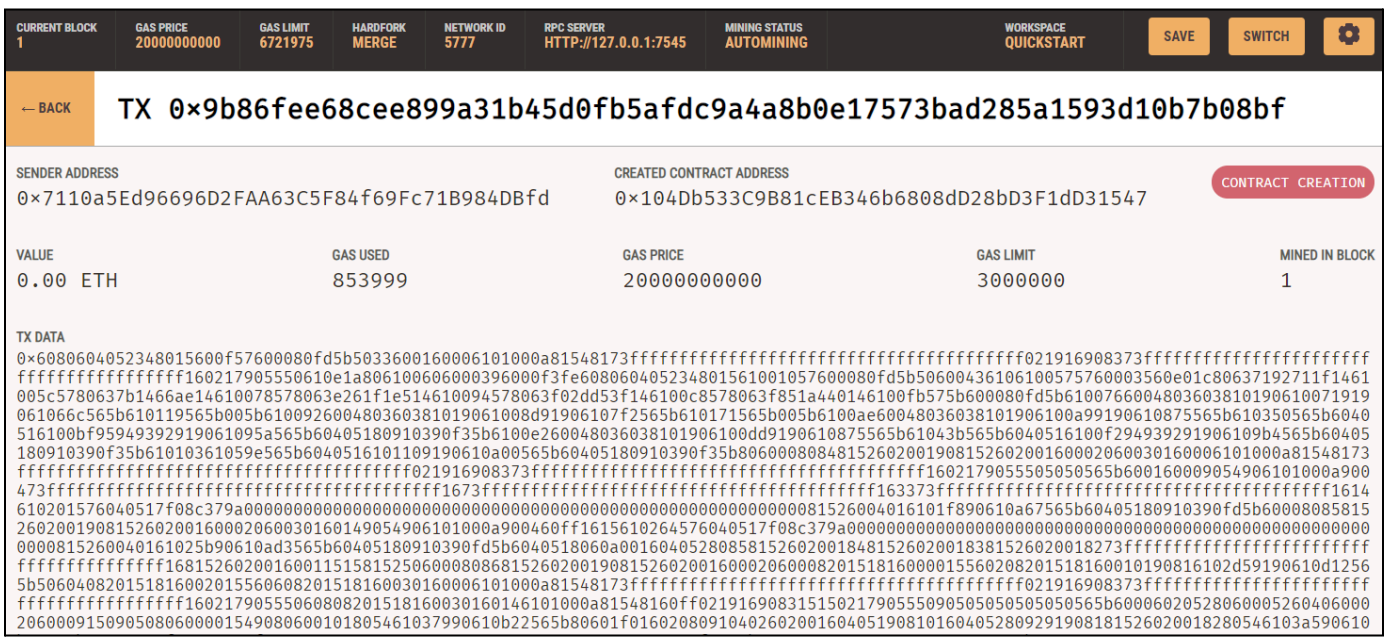




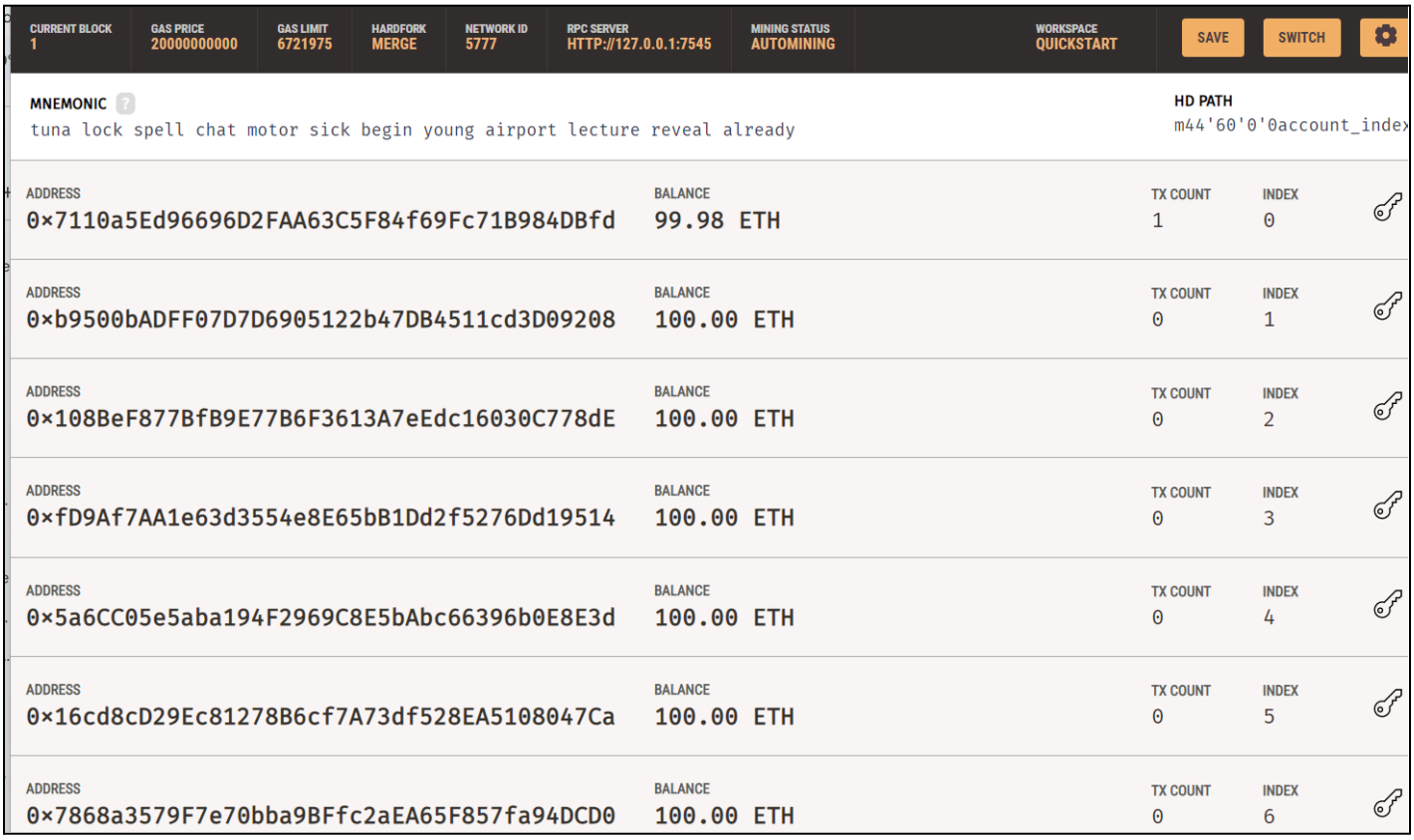
5. Check Transaction Details in Ganache

1. Open the **Ganache Desktop App**.
2. Click the **Transactions** tab at the top.
3. You will see a "Contract Creation" transaction. Click it to see the gas used, the "TX Hash," and the block number





In the **Accounts** tab, we can see that the some balance has been deducted from the first address as part of the transaction that has taken place



Conclusion:-

The experiment successfully demonstrated the deployment of a LandRegistry smart contract on a local blockchain using Ganache, MetaMask, and Remix IDE. By configuring a custom RPC network and aligning the Chain ID, a secure bridge was established to authorize a pre-funded account for transaction gas. This process verified the contract's compilation and its successful migration to a private ledger, confirming the integrity of the decentralized data storage. The setup provides a cost-effective, local environment for testing full-scale blockchain application logic.